

THE VOL SURFACE CONSTRUCTION PLAYBOOK

DESK2QUANT SAMPLE - NOT FOR RESALE

7-PAGE SAMPLE PREVIEW

SVI | SSVI | STATIC ARBITRAGE | DUPIRE LOCAL VOL

Desk2Quant

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This sample is intentionally limited.

The full product includes the 79-page playbook and executable Jupyter lab.

1. THE VOLATILITY SURFACE AS AN OBJECT

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A production volatility surface is more than a table of implied volatilities.

It must fit liquid quotes, interpolate coherently, remain free of static arbitrage, and support stable differentiation for downstream models.

Natural coordinates:

$$k = \log(K/F_T)$$

$$w(k,T) = \sigma_{BS}(k,T)^2 * T$$

Core diagnostics in the full playbook:

- butterfly arbitrage
- calendar arbitrage
- local-vol stability
- repricing closure

2. RAW SVI SMILE PARAMETERIZATION

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Raw SVI represents total implied variance as

$$w(k) = a + b [\rho (k-m) + \sqrt{(k-m)^2 + \sigma^2}]$$

Practical calibration is not just curve fitting.

A desk-quality implementation also checks parameter bounds, total-variance positivity, tail behaviour, and the butterfly functional $g(k)$.

The companion notebook calibrates slices numerically and audits each fit.

3. SSVI SURFACE CONSTRUCTION

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SSVI turns smile slices into a coherent surface.

$$w(k, \theta) = \theta/2 * [1 + \rho * \phi(\theta)^k + \sqrt{(\phi(\theta)^k + \rho)^2 + 1 - \rho^2}]$$

Typical power-law shape:

$$\phi(\theta) = \eta / (\theta^\gamma * (1 + \theta)^{1-\gamma})$$

The full playbook connects the global SSVI parameters to raw-SVI slices and verifies the mapping numerically.

4. STATIC-ARBITRAGE AUDIT

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Butterfly arbitrage can be diagnosed directly from total variance.

$$g(k) = (1 - k \cdot w_k / (2w))^2 - (w_k^2 / 4) \cdot (1/w + 1/4) + w_k k / 2$$

A valid slice requires $g(k) \geq 0$ over the domain used for pricing.

Calendar consistency requires total variance not to decrease with maturity at fixed log-moneyness:

$$d_T w(k, T) \geq 0$$

5. DUPIRE LOCAL VOLATILITY

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Once the surface is smooth and arbitrage-safe, Dupire local variance follows from the time derivative of total variance divided by the butterfly term.

$$\sigma_{\text{loc}}^2(k,T) = w_T(k,T) / g(k,T)$$

This is why surface construction and local-vol extraction cannot be separated: small derivative noise or a near-zero denominator can create explosive local vol.

The notebook includes denominator diagnostics and grid-stability tests.

6. REPRICING CLOSURE AND FAILURE EXPERIMENTS

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A production build should be challenged, not merely plotted.

The companion lab deliberately breaks:

- butterfly constraints
- calendar monotonicity
- near-expiry stability

It then performs Monte Carlo local-vol repricing and compares vanilla prices against Black-Scholes values using standard-error z-scores.

Full product: 79-page PDF + executed notebook + plots + requirements + checksums.